

WASP-104b

R-2354

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_200226 · created 2026-08-19T01:33:42+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458905.8096 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.057 +/- 0.045
Transit depth (Rp/Rs)^2	0.32 +/- 0.51 [%]
Orbital Inclination (inc)	87.97 +/- 2.91
Airmass coefficient 1 (a1)	4.66 +/- 2.75
Airmass coefficient 2 (a2)	-1.43 +/- 1.58
Scatter in the residuals of the lightcurve fit is	56.86 %
Best Comparison Star	#1 - [214, 247]
Optimal Aperture	1.18
Optimal Annulus	5.59
Transit Duration (day)	0.084 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.057 $\pm$ 0.045 vs 0.1214 (deviation 1.42 σ)
Duration fit vs published	0.084 d vs 0.0742 d (deviation 0.81 σ)
Mid-time O-C	-0.1 min
Depth SNR	1.3
Residual scatter	56.86 %

Light curve

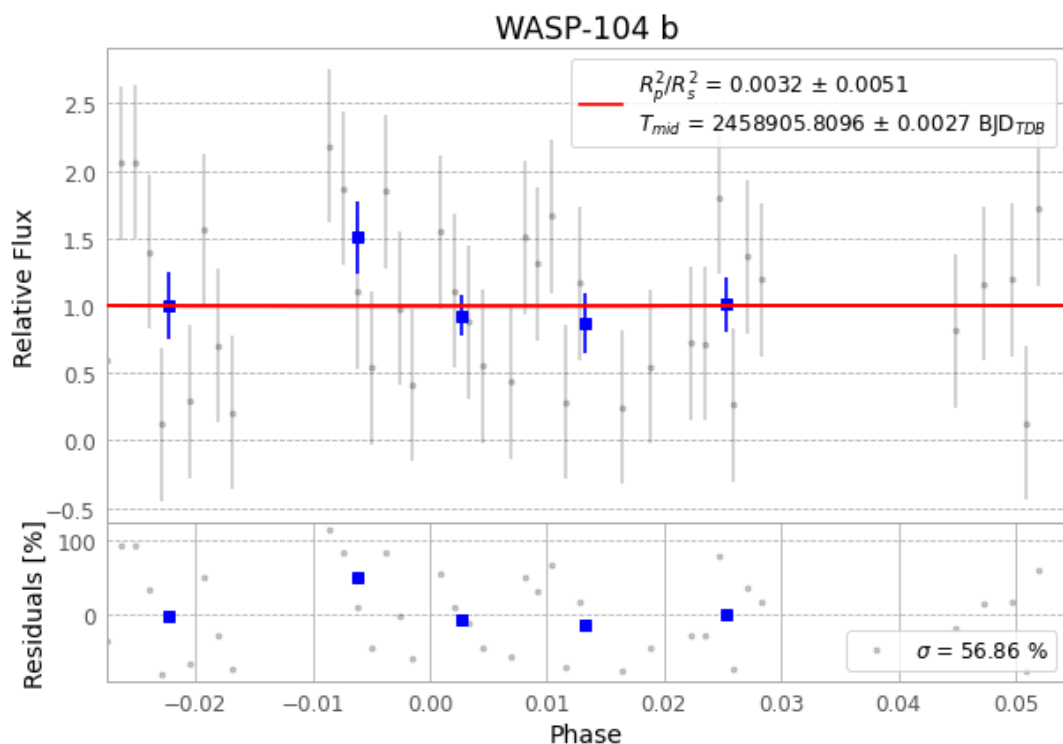


Figure 1 — FinalLightCurve_WASP-104 b_2020-02-25.png (SHA-256 2d4b55d94a528536...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [189, 274], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 4e3954d8a983067e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

71 FITS frames (47 MB), WASP-104200226060612.FITS → WASP-104200226093612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-200226.log (85a2f7a55d35...), FinalLightCurve_WASP-104 b_2020-02-25.png (2d4b55d94a52...), FinalLightCurve_WASP-104 b_2020-02-25.csv (172df4918281...)

Compute resources

Wall time 118.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 01:33Z.